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ECEN 474/704
Lab 3: Prelab
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1. **Minimum transistor dimensions.** What are the absolute minimum dimensions of a transistor? Explain your reasoning, including the minimum contact size. Do we use this minimum size or a slightly larger size for convenience?

   Solution.

   (a) Minimum channel dimensions. For the low-voltage MOS devices used in this lab, the CMHV7SF design manual specifies the following drawn dimensions:

   $$\boxed{L_{\text{min}} = 0.180\,\mu\text{m}, \qquad W_{\text{min,channel}} = 0.220\,\mu\text{m}.}$$

   These are the channel dimensions, rather than the dimensions of the entire contacted transistor layout.

   (b) Contact constraint. A source/drain contact (CA) is $0.200\,\mu\text{m}\times0.200\,\mu\text{m}$, and the minimum active-region (RX) enclosure is $0.100\,\mu\text{m}$ on each side. Thus the diffusion width at a contact must be at least

   $$W_{\text{contact region}} = 0.100 + 0.200 + 0.100 = 0.400\,\mu\text{m}.$$

   If the source, channel, and drain use one constant-width rectangular diffusion strip, the minimum contacted dimensions are therefore

   $$\boxed{W_{\text{min,rectangular}} = 0.400\,\mu\text{m}, \qquad L_{\text{min}} = 0.180\,\mu\text{m}.}$$

   A $0.220\,\mu\text{m}$ channel remains possible with a dogbone-shaped diffusion region: the source/drain contact regions are widened to at least $0.400\,\mu\text{m}$, while the channel stays narrower. The diffusion transitions must also satisfy the gate-to-RX-corner spacing rules. A $0.200\,\mu\text{m}$ contact cannot simply be placed inside a constant-width $0.220\,\mu\text{m}$ strip.

   (c) Practical choice. For convenient contact placement, repeated fingers, and routing, this lab uses a slightly larger finger width:

   $$\boxed{W_f = 0.500\,\mu\text{m}, \qquad L = 0.180\,\mu\text{m}.}$$

   The width is larger than the minimum contacted rectangular width, while the channel length remains at its minimum. The $0.500\,\mu\text{m}$ width is a convenient design choice, not the absolute minimum specified by the PDK.

   Rule references: CMHV7SF Design Manual, applicable to V1.3.1.0, Table 24, p. 67, Rules 1–2 and 10–11; Table 26, p. 77, Rules 200 and 204. The dogbone transition is also subject to Rule 114 on p. 69.

2. **Common-centroid current mirror.** Draw a common-centroid layout of a simple current mirror with equal transistors of $L = 180\,\text{nm}$ and $W = 12\,\mu\text{m}$. Use a finger width of 500 nm, show the drain, source, gate, and bulk connections, include dummy transistors, and provide a floor plan.

Solution.

(a) Finger count and floor plan. The number of fingers in each branch is

$$n_f = \frac{W}{W_f} = \frac{12}{0.50} = 24.$$

There are 48 functional fingers in total. Choose four rows of 12 functional fingers, with one dummy at each end of every row:

$$\boxed{4 \times 12 \text{ functional fingers + 8 dummies.}}$$

Let A denote a reference-branch finger, B an output-branch finger, and X a dummy. The floor plan is

X	A	B	B	A	A	B	B	A	A	B	B	A	X
X	B	A	A	B	B	A	A	B	B	A	A	B	X
X	B	A	A	B	B	A	A	B	B	A	A	B	X
X	A	B	B	A	A	B	B	A	A	B	B	A	X

Each row contains six A and six B fingers, giving 24 of each overall. With equal finger pitch and symmetric row placement, the two groups have the same centroid. Each functional finger has $W_f/L = 0.50/0.18\,\mu\text{m}$; dummies do not contribute to the specified effective transistor width.

(b) Electrical connections. For an NMOS current mirror, use the following nets:

Finger	D	G	S	Bulk
A	I_{IN}	I_{IN}	V_{SS}	V_{SS}
B	I_{OUT}	I_{IN}	V_{SS}	V_{SS}
X	Adjacent D_1 or D_2	V_{SS}	V_{SS}	V_{SS}

The reference branch is diode-connected: $G = D_1 = I_{\text{IN}}$. All functional gates share this node, while $D_2 = I_{\text{OUT}}$. Each dummy has $V_{GS} = 0$ and its gate is isolated from the functional gate bus. Substrate taps connect the bulk to V_{SS}.

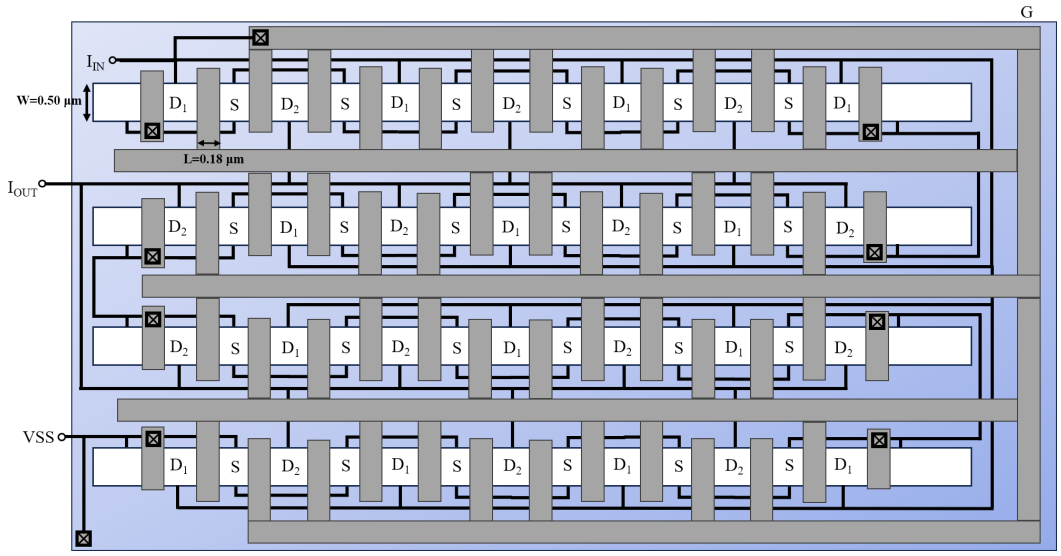


Figure 1: Common-centroid current-mirror layout with four rows, twelve functional fingers per row, and eight edge dummies.

(c) Matching considerations. Use equal finger dimensions and regular spacing, share diffusion where appropriate, and balance the drain routing, contact counts, and parasitic loading between the branches. The outer two rows and inner two rows exchange A and B, balancing the drain regions at the row ends. Common-centroid placement suppresses first-order spatial-gradient mismatch; it does not eliminate random mismatch.

3. **Common-centroid MOM capacitor array.** Draw a common-centroid MOM capacitor array with $C_1 : C_2 = 1 : 1.4$, using eight unit capacitors ($I_1 = I_2 = 4$) and one non-unit capacitor. Each unit measures $10.56\,\mu\text{m} \times 10.56\,\mu\text{m}$. Determine the non-unit dimensions, show the interconnections, and include dummy capacitors and suitable matching techniques.

Solution.

(a) Required non-unit capacitance. Let C_u be the unit capacitance and $C_{\text{nu}} = NC_u$. Then

$$C_1 = 4C_u, \qquad C_2 = (4+N)C_u, \qquad \frac{4}{4+N} = \frac{1}{1.4}.$$

Therefore,

$$\boxed{N = 1.6, \qquad C_{\text{nu}} = 1.6C_u, \qquad C_2 = 5.6C_u.}$$

(b) Non-unit capacitor dimensions.

Using $L_0 = 10.56\,\mu\text{m}$ and the lab manual's area/perimeter matching formulas,

$$\begin{aligned} L_{\text{nu}} &= L_0\left[N + \sqrt{N(N-1)}\right] \\ &= 10.56\left[1.6 + \sqrt{1.6(0.6)}\right] = 27.24264467\,\mu\text{m}, \\ W_{\text{nu}} &= \frac{NL_0^2}{L_{\text{nu}}} = \frac{1.6(10.56)^2}{27.24264467} = 6.54935533\,\mu\text{m}. \end{aligned}$$

Thus,

$$\boxed{W_{\text{nu}} \times L_{\text{nu}} \simeq 6.549\,\mu\text{m} \times 27.243\,\mu\text{m}.}$$

The area and perimeter scale by the same factor:

$$\begin{aligned} A_u &= 111.5136\,\mu\text{m}^2, & P_u &= 42.24\,\mu\text{m}, \\ A_{\text{nu}} &= 178.42176\,\mu\text{m}^2 = 1.6A_u, & P_{\text{nu}} &= 67.584\,\mu\text{m} = 1.6P_u. \end{aligned}$$

Consequently, $P_{\text{nu}}/A_{\text{nu}} = P_u/A_u$. These are the nominal dimensions from the lab's matching construction; the implemented MOM capacitance also depends on the PDK model and available geometry increments.

(c) Common-centroid floor plan. Place the four A units of C_1 at the corners of the inner array, the four B units of C_2 at the side centers, and C_{nu} at the center. Surround the array with unit-size dummies X:

$$\begin{array}{ccccc} X & X & X & X & X \\ X & A_1 & B_1 & A_2 & X \\ X & B_2 & C_{\text{nu}} & B_3 & X \\ X & A_3 & B_4 & A_4 & X \\ X & X & X & X & X \end{array}$$

This gives four A units, four B units, one non-unit capacitor, and sixteen dummies. The table denotes placement, not equal-size physical cells: C_{nu} is a long rectangle, so the spacing must accommodate it symmetrically. Both unit groups have their centroids at the center, and adding the centered C_{nu} preserves the capacitance-weighted centroid of C_2.

(d) Interconnections and matching. Connect A_1–A_4 in parallel and connect B_1–B_4 and C_{nu} in parallel. In particular, C_{nu} belongs only to C_2. All backplate terminals connect to V_{SS}.

Use identical orientation and local routing for the unit capacitors, multiple contacts where appropriate, and balanced routing parasitics. Following Fig. 3-10 of the lab manual, short the two signal terminals of each dummy together as an individual floating node; its backplate remains connected to V_{SS}.

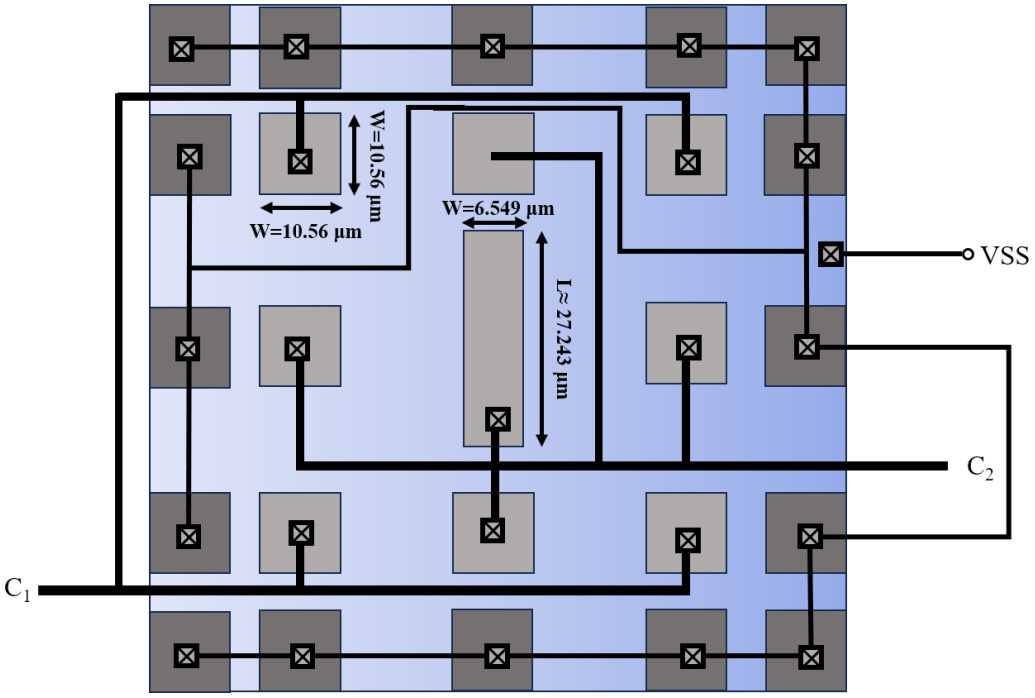


Figure 2: Common-centroid MOM capacitor array. The four C_1 units are connected in parallel; the four C_2 units and the centered non-unit capacitor are connected in parallel.

4. **Interdigitized matched polysilicon resistors.** Draw two matched `oprppresx` resistors of 15 kΩ each, using four series segments per resistor. Account for contact resistance, determine the segment length and width, and use an interdigitized layout. The combined component area of the two resistors must be between 30 and 80 μm^2.

Solution.

(a) Segment dimensions including contact resistance. Each of the four series segments must have nominal resistance

$$R_{\text{seg}} = \frac{15}{4} = 3.75\,\text{k}\Omega.$$

From Table 3-1 of the lab manual,

$$R_s = 0.165\,\text{k}\Omega/\square, \qquad R_{\text{con}} = 0.015\,\text{k}\Omega \cdot \mu\text{m}.$$

The nominal resistance of each contacted segment is

$$R_{\text{seg}} = R_s \frac{L}{W} + \frac{2R_{\text{con}}}{W}.$$

Choose $W = 0.50\,\mu$m. Measuring L in μm and resistance in kΩ gives

$$3.75 = 0.165\frac{L}{0.50} + \frac{2(0.015)}{0.50} = 0.33L + 0.06.$$

Therefore,

$$\boxed{L = 11.1818\,\mu\text{m}, \qquad W = 0.50\,\mu\text{m}.}$$

Each segment contributes 3.69 kΩ from its body and 0.06 kΩ from its two contacts. Hence

$$\boxed{R_1 = R_2 = 4(3.69 + 0.06) = 15.00\,\text{k}\Omega.}$$

The length L is the effective resistor-body length defined by the OP mask, rather than the full strip length including the contact ends.

(b) Area check. The combined body area of the eight functional segments is

$$A_{\text{active}} = 8WL = 8(0.50)(11.1818) \simeq \boxed{44.73\,\mu\text{m}^2},$$

which lies between 30 and 80 μm^2. Including two identical dummy bodies gives approximately 55.91 μm^2. The overall layout footprint is larger because contacts, routing, and spacing require additional area.

(c) Placement and series connections. Place ten parallel strips with equal orientation and spacing:

$$X \quad A_1 \quad B_1 \quad A_2 \quad B_2 \quad A_3 \quad B_3 \quad A_4 \quad B_4 \quad X.$$

The two groups are spatially interdigitized and electrically connected as

$$\text{R1} \longrightarrow A_1 \longrightarrow A_2 \longrightarrow A_3 \longrightarrow A_4 \longrightarrow \text{R1},$$
$$\text{R2} \longrightarrow B_1 \longrightarrow B_2 \longrightarrow B_3 \longrightarrow B_4 \longrightarrow \text{R2}.$$

For a serpentine connection, join the bottom ends of segments 1 and 2, the top ends of segments 2 and 3, and the bottom ends of segments 3 and 4 within each group. The top ends of segments 1 and 4 are the external terminals. This is an interdigitized arrangement, as requested; the single-row $ABABABAB$ pattern is not common-centroid.

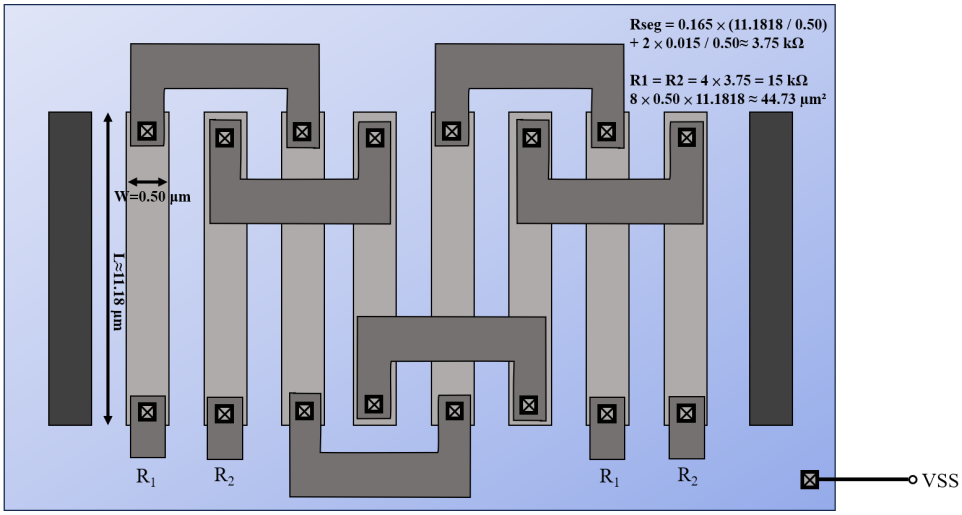


Figure 3: Interdigitized matched resistor pair with four series segments per resistor and one dummy strip at each outer edge.

(d) Matching and substrate connections. Keep segment geometry, contact counts, routing lengths, and the lengths of metal overlapping the resistor bodies balanced between the two groups. Where separate nets cross, use different metal layers without an interconnecting via, while balancing the layer usage and via counts between the groups.

Following Fig. 3-14 of the lab manual, leave each dummy's two signal terminals individually floating. All resistor backplates, including those of the dummies, connect to V_{SS}.